

KSG Revision-4 M1a Composite-v4 Evidence Process

An append-only repair for an uninstantiable hosted-evidence contract

Sepehr Mahmoudian

15 August 2026

Abstract

This report specifies the bounded, append-only migration from an internally inconsistent composite-v3 hosted-evidence contract to composite-v4. It records two separately sufficient counterexamples, defines an acyclic two-commit topology, retains raw hosted response and artifact bytes before deriving a typed receipt, and separates byte custody from every mathematical, estimator, application, and scientific-novelty claim. The frozen v3 authorities remain historical facts; they are neither rewritten nor presented as having issued a receipt.

Disposition. This document specifies project-defined process evidence. It is not a PID definition, a KSG proof, estimator validation, peer review, authentication, attestation, or application evidence. The frozen composite-v3 files remain historical authorities; no v3 receipt is issued or retroactively repaired.

1 Why a new contract is necessary

The recovery subject is Git commit `bc3aa80fb6025e709c2906a08bce25a4fac40578`, tree `7d87f87953a42edb91e40880d918471c7cbe4414`. Its hosted CI and CodeQL runs succeeded. That success does not make every possible receipt contract satisfiable. Composite-v3 has two separately sufficient contradictions for this exact subject.

1.1 Opaque analysis identifiers cannot satisfy two orders

V3 requires four CodeQL rows in semantic language order

(actions, javascript-typescript, python, rust)

and also requires their provider analysis identifiers to be strictly increasing. The observed IDs in that language order are

(1617732991, 1617732745, 1617735963, 1617735749).

They are positive and unique but not increasing. Sorting the rows would violate the language positions; moving or inventing identifiers would falsify the observation. Provider identifiers are opaque keys, not scientific quantities. Composite-v4 therefore retains one semantic order and requires uniqueness only within the analysis-ID resource type.

GitHub’s code-scanning alert endpoints expose repository-level current state, not alerts foreign-keyed to a workflow run and not a reconstruction of alert state during that run. The recovery and C4 role labels group when those endpoint bytes were captured; they do not create a run-to-alert relation. V4 deliberately requires the resulting current-state partition to equal the retained 191-alert baseline as an operational admission condition. That equality is neither a causal attribution to either run nor evidence about PID mathematics.

1.2 Three additions require a fourth source-state change

The frozen v3 direct-child rule permits exactly three additions: one receipt and two historical index payloads. The repository-wide source-state manifest inventories every repository-visible path except itself. Adding three paths changes the required self-excluding projection, so a truthful child must also modify the manifest. A three-addition tree is stale; a four-path tree violates v3. This contradiction does not rely on the CodeQL-order contradiction.

The machine-readable witness is `audit/evidence/ksg-rev4-m1a-composite-v3-impossibility-2026-08-15.json`. It binds the frozen v3 checker, self-test, schema, current-source checker and manifest, and CI workflow by Git blob ID, SHA-256, and byte count.

2 Evidence ontology and custody boundary

The durable subjects are Git commit, tree, and blob bytes; canonical manifests; exact HTTPS response bodies; and downloaded artifact archives. A Git index is mutable local implementation state. Two historical index hashes and sizes remain observations, but their exact bytes are no longer available to this process. V4 neither reconstructs them nor substitutes a different index.

Object	What is bound	What is not implied
Git object	Raw bytes rehashed under Git SHA-1; exact parent, tree, mode, and path	Repository origin, trust, authorship, or review
Source manifest	Full containing-tree projection except the manifest itself	Its own commit, authentication, or atomic snapshot
Hosted response	Two bounded response bodies, request/page identity, SHA-256, byte count	Provider authenticity, wall-clock completeness, response order
Artifact ZIP	API digest, exact archive bytes, safe member paths, member hashes	Semantic correctness of member content beyond named checks
Typed receipt	Deterministic projection re-derived from raw captures	Scientific validity, estimator quality, release readiness

Table 1: V4 separates byte custody from scientific interpretation.

3 Acyclic two-commit topology

3.1 C4: contract migration

C4 is an unsigned, single-parent direct child of the recovery commit. It contains the v4 checker, self-test, capture tool, closed schemas, dedicated workflow, path policy, impossibility witness, this TeX source and PDF, bounded operational custody gates, schemas, documentation, the

prospective research-integrity workflow publication, and a separately scoped current-project Lean replay/reseal. No KSG/PID theorem statement, scientific-object definition, numerical result, or disposition changes, and composite-v4 derives no KSG/PID credit from that replay. C4 binds the authored research-method bytes, but neither the v4 checker, hosted runs, nor Lean replay adjudicates the external-source observations or proves protocol adoption. The self-excluding source manifest is generated last. C4 contains neither hosted capture nor typed v4 receipt and therefore makes no prospective hosted-success claim.

3.2 R4: observation receipt

After exact-C4 hosted runs are terminal, R4 is an unsigned, single-parent direct child of C4 with exactly three source changes:

1. add the canonical byte-retaining hosted capture;
2. add the canonical typed receipt derived from that capture; and
3. modify the self-excluding source manifest so it inventories both additions.

The receipt hashes the capture but not itself, the regenerated manifest, R4's tree, or R4's commit. Those identities are derived from Git. This orientation removes the checksum cycle while keeping both evidence objects inside source-state custody.

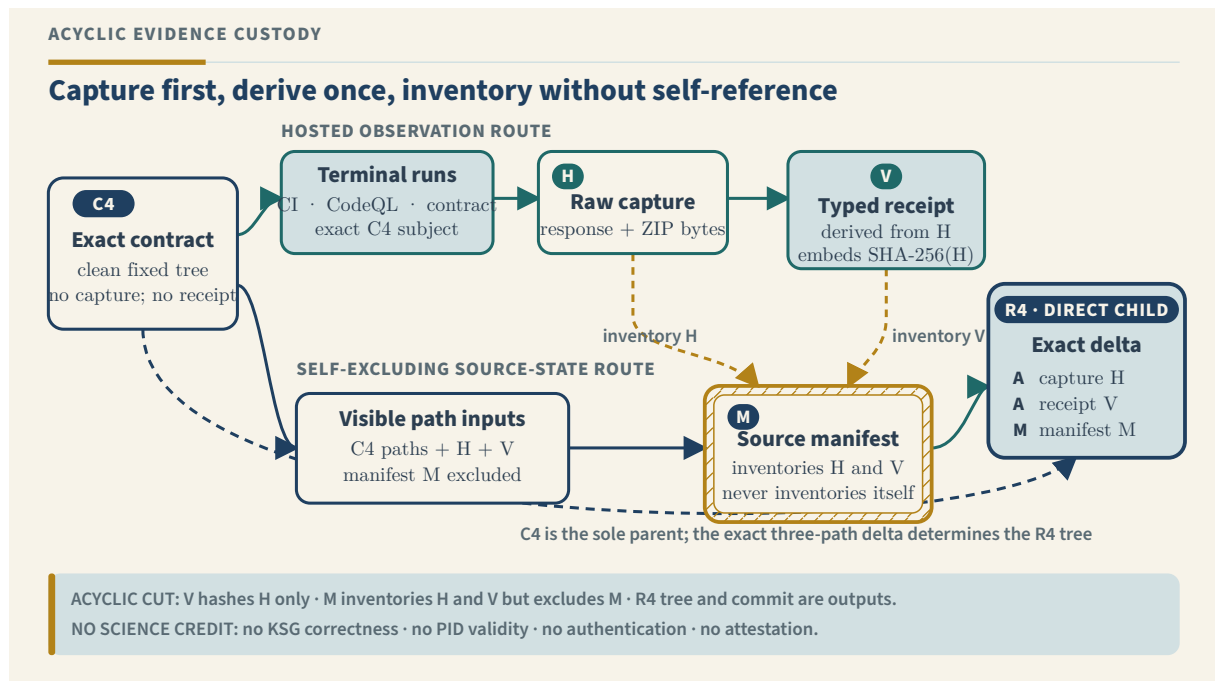


Figure 1: Acyclic C4-to-R4 byte custody. The typed receipt V is derived from the raw capture H and embeds the SHA-256 digest of H. The self-excluding manifest M inventories H and V, but never itself. R4 is the unsigned, single-parent direct child of C4; its exact tree delta adds H, adds V, and modifies M. The R4 tree and commit identities are outputs, never hash inputs. These arrows establish byte dependencies only—not KSG correctness, PID validity, authentication, attestation, or scientific credit.

3.3 Rejected local pre-publication replay attempt

Before any C4 push, the first one-shot local r9 execution and its receipt-finalization predicates passed. The emitted canonical schema-v2 document has internal `status = passed`; this is bounded execution-custody evidence only. The containing C4 static check then rejected publication. For the same 748 sorted entries, the current-source generator and first composite checker disagreed by exactly one terminal line feed:

Generator, newline-free compact JSON: 132,893 bytes.

SHA-256: b336e6f54450090693731f2391b1ef3e112095dd9a9c8cbdaddbdf2f855fba47.

First composite checker, same bytes plus one terminal line feed: 132,894 bytes.

SHA-256: 7d4d4fd6bc478aeb20008cd05d1efe4b92b6ca9fd72a72043e67322e6a722f20.

Local commit e02d27bec91f142949336f9f28550c672d22b297 was never pushed or accepted as C4. The candidate receives no C4-publication, hosted, scientific, accepted-current-replay, or independence credit. Its exact 132,710 bytes are retained unchanged:

Artifact: audit/evidence/lean-4.33.0-darwin-aarch64-current-project-replay-2
026-08-15-r9-prepublication-closure-rejected-2026-08-17.json.

SHA-256: fb162cc40da3059b61eab9024f4aa38cf6daf2d84ef7e1d8a26dc7d345291e70.

The corrected checker adopts the generator's already-published newline-free compact framing and requires a fresh one-shot replay. The accepted r9 slot may be filled only by a subsequent fresh attempt over the complete corrected static-and-hostile surface, and it must bind the rejected bytes as retained pre-publication closure evidence outside the accepted replay lineage; the rejected document is neither renumbered nor relabelled as the current receipt. This publication preserves the exact rejected replay and measured framing facts; it does not represent the rejected local C4 object as a published commit.

3.3.1 Withdrawn replay after a stale hostile fixture

After the compact-framing repair, a fresh local replay ran from 2026-08-17T21:20:16.677815Z through 2026-08-17T21:36:37.989770Z and emitted a 132,912-byte canonical schema-v2 document with internal `status = passed` and SHA-256 6d5068a2ade251b4ea005e847b78be656bb7697b14ae2e6d8a644d521f09e2cb (local Git blob 3d83a833401160f8e980690b0f5d11cef5dfca19). It bound composite checker SHA-256 8640bbe11fa37011921291c719fd50bb02ac1baaf945219a4e723f46ddc7d106. The reported local Lean execution, receipt finalization, 114-mutation suite, and composite static checker in normal and optimized modes passed. Before any push, the complete composite self-test stopped publication: its 3f965f442fa925cd17aa466f5fee3b24207c3d2eaf98b74801c2102d48563642 fixture incorrectly classified identical ZIP bytes delivered once directly and once through an individually safe redirect as inconsistent. The schema and capture tool permit either delivery per request; the raw capture hash retains the route, while repetition equality compares normalized provider semantics.

The corrected self-test, SHA-256 4a40f43396e5b45aa0a5762995eabb908b61d850b1a215aa817c731140a6078b, uses mixed direct/safe redirect delivery as a positive and rejects a forbidden redirect host. It also makes the repetition and duplicate-response mutations reach their named predicates rather than an earlier workflow-name or malformed-ZIP check. Host-local commit f6d76c1d01a040f74ea55277a5ba835b32fdb6ab with tree f5c306ab982d84d84fa47a9b4f38407bc0f843a4 and parent bc3aa80fb6025e709c2906a08bce25a4fac40578 was never pushed or accepted as C4; those local object identities are neither a durable locator nor authentication. The candidate was not promoted or selected and receives no C4-publication, hosted, scientific, accepted-current-replay, selection, or independence credit. Reuse or relabelling is forbidden.

C4 retains this exact identity, disposition, and omission record, but deliberately omits the second raw preimage; its digest alone provides neither recovery nor reproducibility. Unlike the first retained rejected document, whose byte framing is the decisive closure witness, this second document neither caused nor diagnosed the failure; the stale fixture and corrected causal negative are the decisive artifacts. Reopening requires frozen corrected bytes, complete normal/optimized static and hostile suite success, a fresh one-shot replay, current-source regeneration last, and a newly validated exact C4.

4 Bounded hosted-capture protocol

Every load-bearing logical response is fetched twice. Each retained row contains the exact initial API path, page, repetition, response kind, response SHA-256, byte count, canonical base64 body, and redirect observation. Pagination continues through one explicit empty terminal page. Duplicate request identities, duplicate provider resource IDs, missing pages, early empty pages, and unequal normalized repetitions fail closed.

The bounded request set covers:

- recovery and C4 CI run, complete jobs/steps, artifact listings, and every artifact ZIP;
- recovery and C4 CodeQL run, complete jobs, exact-head analyses, and all open, dismissed, and fixed alert pages; and
- the dedicated composite-v4 workflow run, jobs/steps, and exact static-validation ZIP.

Artifact redirects are handled explicitly. The initial authenticated request may redirect only to a reviewed HTTPS artifact-host suffix. The authorization header is stripped before the second request; ambient proxies are disabled. The record retains the redirect status, target host, and SHA-256 of the full target URL, never the signed target URL itself.

Retry is narrow and observable: at most two retries for HTTP 429, 502, 503, 504, or a bounded transport failure. Authentication, authorization, semantic, JSON, pagination, duplication, and cardinality failures are never retried. A failed attempt contributes no observation credit.

4.1 Executable C4-to-R4 route

Run the following only from an exact clean C4 after all three migration runs are terminal. The caller has already opened the GitHub token read-only on descriptor 3. Exact C4, tree, and run IDs are supplied explicitly. The temporary parent must resolve outside the repository. A nonzero producer exit deletes its partial standard output.

```
set -eu
repo_root=$(pwd -P)
: "${C4:?set exact C4 commit}"
: "${C4_TREE:?set exact C4 tree}"
: "${MIGRATION_CI_RUN:?set exact C4 CI run}"
: "${MIGRATION_CODEQL_RUN:?set exact C4 CodeQL run}"
: "${MIGRATION_CONTRACT_RUN:?set exact dedicated-workflow run}"
temp_parent=$(cd "${TMPDIR:-/tmp}" && pwd -P)
case "$temp_parent/" in "$repo_root/*") exit 2 ;; esac
capture_root=$(mktemp -d "$temp_parent/pid-rs-ksg-v4-capture.XXXXXX")
chmod 700 "$capture_root"; umask 077
capture_tmp="$capture_root/capture.json.tmp"
capture="$capture_root/capture.json"
if python3 -I -S -B scripts/capture-ksg-m1a-composite-v4.py \
```

```

--contract-commit "$C4" --contract-tree "$C4_TREE" \
--migration-ci-run "$MIGRATION_CI_RUN" \
--migration-codeql-run "$MIGRATION_CODEQL_RUN" \
--migration-contract-run "$MIGRATION_CONTRACT_RUN" --token-fd 3 \
>"$capture_tmp"; then mv "$capture_tmp" "$capture"
else status=$?; rm -f -- "$capture_tmp"; exit "$status"; fi
exec 3<&-
receipt_tmp="$capture_root/receipt.json.tmp"
receipt="$capture_root/receipt.json"
if python3 -I -S -B scripts/check-ksg-m1a-composite-v4.py \
--derive-receipt <"$capture" >"$receipt_tmp"; then
mv "$receipt_tmp" "$receipt"
else status=$?; rm -f -- "$receipt_tmp"; exit "$status"; fi
install -m 0644 "$capture" \
audit/evidence/.ksg-rev4-m1a-composite-hosted-capture-v4-2026-08-15.json.tmp
mv audit/evidence/.ksg-rev4-m1a-composite-hosted-capture-\
v4-2026-08-15.json.tmp \
audit/evidence/ksg-rev4-m1a-composite-hosted-capture-v4-2026-08-15.json
install -m 0644 "$receipt" \
audit/evidence/.ksg-rev4-m1a-composite-receipt-v4-2026-08-15.json.tmp
mv audit/evidence/.ksg-rev4-m1a-composite-receipt-v4-2026-08-15.json.tmp \
audit/evidence/ksg-rev4-m1a-composite-receipt-v4-2026-08-15.json
source_state_tmp="$capture_root/current-source-state-v1.json.tmp"
if python3 -I -S -B scripts/check-current-source-state-v1.py --emit \
>"$source_state_tmp"; then
install -m 0644 "$source_state_tmp" \
audit/evidence/.current-source-state-v1.json.tmp
else status=$?; rm -f -- "$source_state_tmp"; exit "$status"; fi
mv audit/evidence/.current-source-state-v1.json.tmp \
audit/evidence/current-source-state-v1.json

```

The result must have exactly the three declared R4 paths. After the unsigned single-parent R4 with exact message Record KSG M1a composite v4 receipt, validate its committed receipt in normal and optimized modes:

```

python3 -I -S -B scripts/check-ksg-m1a-composite-v4.py --validate-receipt \
<audit/evidence/ksg-rev4-m1a-composite-receipt-v4-2026-08-15.json
python3 -O -I -S -B scripts/check-ksg-m1a-composite-v4.py --validate-receipt \
<audit/evidence/ksg-rev4-m1a-composite-receipt-v4-2026-08-15.json

```

The temporary directory is not evidence. Remove it only after the committed blobs and validation receipts are separately identified. A failed or interrupted route receives no hosted-observation credit.

5 Validation stack

1. **Strict representation:** outer capture, receipt, counterexample, policy, and source-state artifacts use exact canonical serialization. Retained provider JSON remains byte-for-byte and is parsed as strict UTF-8 with duplicate-key, floating-point, non-finite, nesting, and integer-width rejection; provider formatting is not claimed canonical.

2. **Closed schema:** exact root and nested key inventories, exact JSON scalar types, and full semantic checks beyond what portable JSON Schema can express.
3. **Raw-to-typed join:** each raw repetition is parsed separately; run, job, step, artifact, analysis, and alert projections must agree exactly.
4. **Provider-resource joins:** every job binds its run, attempt, head SHA, terminal state, and UTC interval; every analysis records only a fixed language/category and job-name match, not a provider-supplied analysis-to-job foreign key.
5. **Artifact join:** archive API digest, downloaded bytes, safe ZIP members, and post-commit source-state fields are re-derived against the committed tree.
6. **Git graph:** raw commit, tree, and blob bytes are rehashed in process separately from Git-reported object IDs; C4/R4 parentage, messages, modes, and path deltas are exact. In every checked post-R4 descendant, exact byte retention applies to the v4-authority paths explicitly enumerated by the checker. Other C4 paths remain historically bound by the C4 tree and r9 map; a later version must be recorded by a truthfully regenerated current-source manifest rather than being mistaken for the original bytes.
7. **Source state:** the complete deterministic manifest is rebuilt in process from the containing tree, including critical artifacts, PDFs, subprojections, and nonclaims.
8. **Mutation and runtime parity:** named representative load-bearing fields receive type-correct, inconsistent, and unauthorized-field reseals. The exact mutation inventory, not this category label, defines the exercised coverage. Mixed direct and individually safe redirected delivery of identical ZIP bytes is a positive transport variation; an unsafe redirect host is the causal negative. The dedicated hosted workflow requires byte-identical normal and optimized Python 3.11 dispositions; broader project minor-version coverage is a separate CI surface and is not implied by this receipt.

These predicates cannot detect a fully coordinated fabricated capture and matching receipt that satisfy every declared condition; that would require a separately authenticated external anchor. The checker therefore validates retained bytes and their deterministic interpretation without authenticating their origin.

6 PID and method firewall

This contract concerns custody for one KSG M1a implementation lineage. Similar terminology does not merge scientific objects. Each downstream claim must name its definition, estimand, support, sample role, units, implementation route, and evidence class.

Model/agent suggestions, proof candidates, model-labelled judge outputs, and generated certificates receive no scientific credit in C4/R4. Research/scientific or claim-adjudicating attempts opened after this policy revision must satisfy the prospective candidate/judge controls in `AGENTS.md` through a pre-candidate object-specific task packet and an append-only attempt ledger covering failed and unselected attempts as well as selected successes. C4/R4 contains no such scientific packet and makes no claim that earlier proofs or certificates complied with that protocol.

Scientific object	Object type	No-transfer boundary
Categorical MGW shared exclusions	Probability-law functional; empirical rows induce a plug-in law	Not Ehrlich continuous PID, not $I_{\min}$, not a generic “Wibral PID”
Schick-Poland construction	General paper construction	Requires its own definition-to-route mapping
Ehrlich continuous shared exclusions	Analytic continuous functional	Distinct from its source-disjunction kNN estimator
KSG	Continuous MI estimator family	MI evidence alone does not validate any PID atom construction
PID2/PID3 atoms	Algebraic compositions under named redundancy terms	Algebra does not choose the redundancy functional or support contract
Williams–Beer $I_{\min}$	Distinct discrete PID	Never a fallback for MGW or Ehrlich routes
Fitted quantized SxPID	Project composition for a quantized estimand	Quantization changes the estimand; no continuous fallback claim
Infomorphic objective	Objective composed from named PID atoms	Not another PID definition; requires an explicit atom-to-objective mapping

Table 2: Green process evidence cannot cross a scientific-object boundary.

7 Research-queue non-evidence

Deferral is not scientific rejection, but this KSG custody migration does not preserve or validate unrelated dirty-overlay payloads. A separate read-only work-item inventory reported a PID2 revision-3 exact-binary64/oracle candidate and a categorical SxPID3 Program-A specification/-counterexample candidate. Their bytes are not retained by C4 or R4, so reported fixture counts and proposed obligations are not preservation evidence here. They remain explicit future research items until an append-only, hash-bound intake records the exact payload, provenance, symbol mapping, open obligations, and an explicit **absent** or **unknown** disposition for any historical pre-candidate packet, judge, attempt ledger, and selection rule. That intake grants no retroactive compliance credit. Every new attempt requires a newly frozen task packet, pre-bound judge/verifier, append-only attempt ledger, and separately scoped validation with declared independence dimensions. Those bytes are absent from C4/R4. Neither candidate may be merged into KSG evidence or used to transfer a result between PID definitions or estimators.

8 Reproduction and publication checklist

1. Rehash the exact recovery commit/tree and every frozen v3 authority.
2. Run checker schema, filesystem, JSON, Git, and synthetic hosted-projection mutation suites under normal and optimized isolated Python. Exercise capture-tool transport security through its separate offline controls; do not infer a network observation from a synthetic fixture.
3. Build this PDF with shell escape disabled; reject warnings; render every page; inspect clipping, overlap, glyphs, links, table width, and page order.
4. Generate the next Lean replay only after every operational byte is quiescent and the generator proves the unique zero projection placeholder, both identical final composite-checker cuts, and the normalized three-placeholder Lean digest before any replay command; retain prior replay receipts append-only and validate the projection/final-custody cuts.
5. Generate current-source last; deterministically reconstruct its complete containing-tree projection and every subprojection.

6. Inspect the exact unsigned C4 object before fast-forward push. Require exact-SHA CI, CodeQL, and dedicated-v4 workflow success.
7. Require the dedicated hosted job to contain exactly one static-validation step, one adversarial self-test step, and one static-artifact upload step; provider-injected setup steps are not substitutes for these user steps.
8. Capture each named hosted response twice. Derive, validate, and inspect R4 before its own unsigned fast-forward push and exact-SHA hosted monitoring.
9. Keep KSG revision-4 `integration_no_go` until the separate M1c scientific packet closes its mathematical, implementation, numerical, statistical, and application gates.

9 Explicit nonclaims

No v3 receipt is issued in the checked lineage; its reserved path remains absent from C4/R4. No historical index payload is reconstructed. No response authenticates itself. No successful workflow is a theorem, estimator validation, peer review, release, or novelty claim. A separate model call or model-labelled judge is not independent review. Composite-v4 does not claim that earlier proofs, certificates, or model runs followed the prospective candidate/judge protocol. No PID result transfers across categorical, continuous, quantized, mixed-support, two-source, three-source, pointwise, averaged, evaluator, estimator, or objective boundaries.

Bound artifact directory: `audit/evidence/`.

Companion source: `ksg-rev4-m1a-composite-v4-process-2026-08-15.md`; **machine counterexample:** `ksg-rev4-m1a-composite-v3-impossibility-2026-08-15.json`.